

ADVANCED

Stage 6

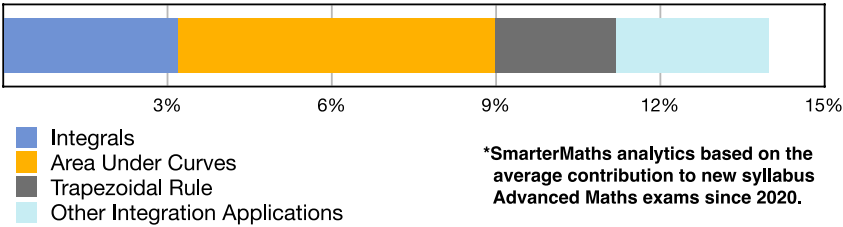
Calculus (Adv), C4 Integration (Adv)

Other Integration Applications (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 132 minutes (based on allocation of 1.5 minutes per mark)

C4 Integration



*SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.

HISTORICAL CONTRIBUTION

- C4 Integration* has contributed a substantial 14.0% to new syllabus Advanced exams since it was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: 1-Integrals (3.2%), 2-Areas Under Curves (5.8%), 3-Trapezoidal Rule (2.2%) and 4-Other Integration Examples (2.8%).
- This analysis looks at *Other Integration Applications*.

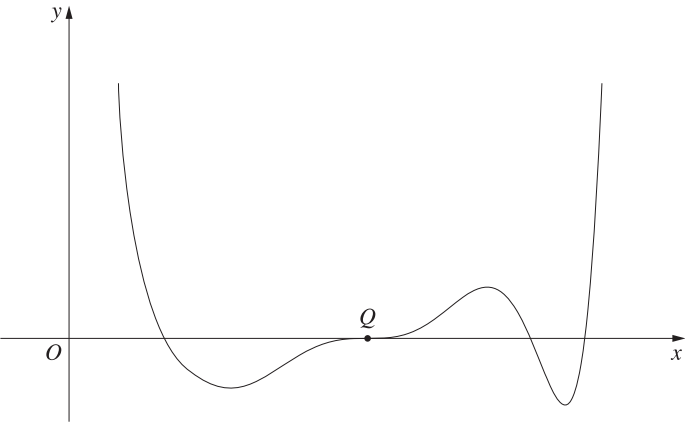
HSC ANALYSIS - What to expect and common pitfalls

- Other Integration Applications* has been examined as a feast or famine in new syllabus exams to date. It contributed to multiple questions in the 2024, 2023 and 2021 exams, and was omitted in 2022.
- Database questions look at similar content to *C3 Rates of Change*, but are allocated in this topic area as they include integration.
- The most common underlying themes are motion, flow (examined in 2024) and population which are all well represented in the database.
- Questions often provide the gradient/primitive function of a curve and require students to find the equation of the curve. Most recently examined with a 4-mark allocation in *2023 Adv 28*, producing a 51% mean mark.
- Some *EQ-Bank* questions in the database take the form of shorter questions that we expect to feature in the new exam format, consistent with NESAs sample exam.

Questions

1. Calculus, 2ADV C4 2024 HSC 10 MC

The diagram shows the graph $y = f(x)$.



The point Q is a horizontal point of inflection.

Let $A(x) = \int_0^x f(t) \, dt$.

How many points of inflection does the graph $y = A(x)$ have?

- A. 2
- B. 3
- C. 4
- D. 5

2. Calculus, 2ADV C4 SM-Bank 1 MC

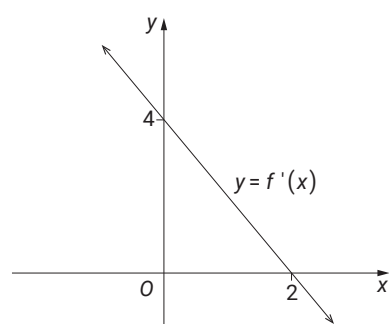
A lift accelerates from rest at a constant rate until it reaches a speed of 3 ms^{-1} . It continues at this speed for 10 seconds and then decelerates at a constant rate before coming to rest. The total travel time for the lift is 30 seconds.

The total distance, in metres, travelled by the lift is

- A. 45
- B. 60
- C. 75
- D. 90

3. Calculus, 2ADV C3 2017 HSC 9 MC

The graph of $y = f'(x)$ is shown.



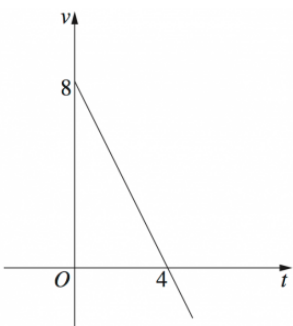
The curve $y = f(x)$ has a maximum value of 12.

What is the equation of the curve $y = f(x)$?

- A. $y = x^2 - 4x + 12$
- B. $y = 4 + 4x - x^2$
- C. $y = 8 + 4x - x^2$
- D. $y = x^2 - 4x + 16$

4. Calculus, 2ADV C4 2015 HSC 9 MC

A particle is moving along the x -axis. The graph shows its velocity v metres per second at time t seconds.



When $t = 0$ the displacement x is equal to 2 metres.

What is the maximum value of the displacement x ?

- A. 8 m
- B. 14 m
- C. 16 m
- D. 18 m

5. Calculus, 2ADV C4 2024 HSC 15

Initially there are 350 litres of water in a tank. Water starts flowing into the tank.

The rate of increase of the volume V of water in litres is given by $\frac{dV}{dt} = 300 - 7.5t$, where t is the time in hours.

Find the volume of water in the tank when $\frac{dV}{dt} = 0$. (3 marks)

6. Calculus, 2ADV C4 2017 HSC 13d

The rate at which water flows into a tank is given by

$$\frac{dV}{dt} = \frac{2t}{1 + t^2},$$

where V is the volume of water in the tank in litres and t is the time in seconds.

Initially the tank is empty.

Find the exact amount of water in the tank after 10 seconds. (3 marks)

7. Calculus, 2ADV C4 2013 HSC 16a

The derivative of a function $f(x)$ is $f'(x) = 4x - 3$. The line $y = 5x - 7$ is tangent to the graph $f(x)$.

Find the function $f(x)$. (3 marks)

8. Calculus, 2ADV C4 2019 HSC 14a

A particle is moving along a straight line. The particle is initially at rest. The acceleration of the particle at time t seconds is given by $a = e^{2t} - 4$, where $t \geq 0$.

Find an expression, in terms of t , for the velocity of the particle. (2 marks)

9. Calculus, 2ADV C4 2023 HSC 13

Let $P(t)$ be a function such that $\frac{dP}{dt} = 3000e^{2t}$.

When $t = 0$, $P = 4000$.

Find an expression for $P(t)$. (2 marks)

10. Calculus, 2ADV C4 EQ-Bank 2

The population, D , of Tasmanian Devils in a sanctuary is given by $D(t)$, where t is the time in years after the sanctuary was established.

The devil population changes at a rate modelled by the function $\frac{dD}{dt} = 28e^{0.35t}$.

Calculate the increase in the number of Tasmanian Devils at the end of the first 8 years. Give your answer correct to three significant figures. (3 marks)

11. Calculus, 2ADV C4 2011 HSC 4c

The gradient of a curve is given by $\frac{dy}{dx} = 6x - 2$. The curve passes through the point $(-1, 4)$.

What is the equation of the curve? **(2 marks)**

12. Calculus, 2ADV C4 2014 HSC 11f

The gradient function of a curve $y = f(x)$ is given by $f'(x) = 4x - 5$. The curve passes through the point $(2, 3)$.

Find the equation of the curve. **(2 marks)**

13. Calculus, 2ADV C4 2008 HSC 5a

The gradient of a curve is given by $\frac{dy}{dx} = 1 - 6\sin 3x$. The curve passes through the point $(0, 7)$.

What is the equation of the curve? **(3 marks)**

14. Calculus, 2ADV C4 2016 HSC 16a

A particle moves in a straight line. Its velocity $v \text{ ms}^{-1}$ at time t seconds is given by

$$v = 2 - \frac{4}{t+1}.$$

- Find the initial velocity. **(1 mark)**
 - Find the acceleration of the particle when the particle is stationary. **(2 marks)**
 - By considering the behaviour of v for large t , sketch a graph of v against t for $t \geq 0$, showing any intercepts. **(2 marks)**
 - Find the exact distance travelled by the particle in the first 7 seconds. **(3 marks)**
-

15. Calculus, 2ADV C4 2009 HSC 7a

The acceleration of a particle is given by

$$a = 8e^{-2t} + 3e^{-t},$$

where x is the displacement in metres and t is the time in seconds.

Initially its velocity is -6 ms^{-1} and its displacement is 5 m.

- Show that the displacement of the particle is given by

$$x = 2e^{-2t} + 3e^{-t} + t. \quad \textbf{(2 marks)}$$

- Find the time when the particle comes to rest. **(3 marks)**
 - Find the displacement when the particle comes to rest. **(1 mark)**
-

16. Calculus, 2ADV C4 2012 HSC 15b

The velocity of a particle is given by

$$v = 1 - 2\cos t,$$

where x is the displacement in metres and t is the time in seconds. Initially the particle is 3 m to the right of the origin.

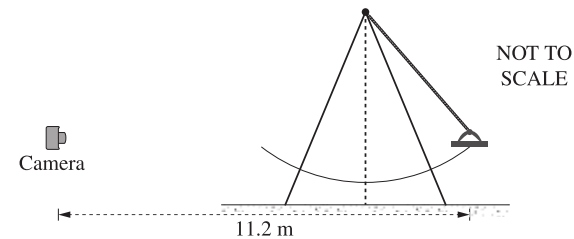
- Find the initial velocity of the particle. **(1 mark)**
 - Find the maximum velocity of the particle. **(1 mark)**
 - Find the displacement, x , of the particle in terms of t . **(2 marks)**
 - Find the position of the particle when it is at rest for the first time. **(2 marks)**
-

17. Calculus, 2ADV C4 2023 HSC 26

A camera films the motion of a swing in a park.

Let $x(t)$ be the horizontal distance, in metres, from the camera to the seat of the swing at t seconds.

The seat is released from rest at a horizontal distance of 11.2 m from the camera.



- The rate of change of x can be modelled by the equation

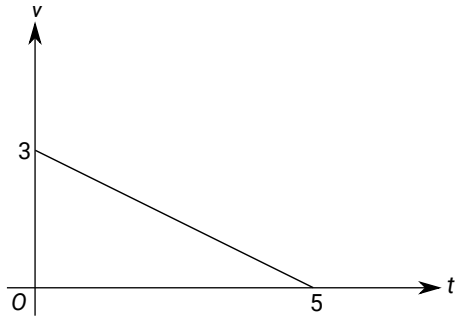
$$\frac{dx}{dt} = -1.5\pi \sin\left(\frac{5\pi}{4}t\right).$$

Find an expression for $x(t)$. **(2 marks)**

- How many times does the swing reach the closest point to the camera during the first 10 seconds? **(2 marks)**
-

18. Calculus, 2ADV C4 EQ-Bank 5

The velocity of a particle moving along the x -axis at v metres per second at t seconds, is shown in the graph below.



Initially, the displacement x is equal to 12 metres.

- Write an equation that describes the displacement, x , at time t seconds. **(2 marks)**
- Draw a graph that shows the displacement of the particle, x metres from the origin, at a time t seconds between $t = 0$ and $t = 5$. Label the coordinates of the endpoints of your graph. **(2 marks)**

19. Calculus, 2ADV C4 2015 HSC 15c

Water is flowing in and out of a rock pool. The volume of water in the pool at time t hours is V litres. The rate of change of the volume is given by

$$\frac{dV}{dt} = 80\sin(0.5t)$$

At time $t = 0$, the volume of water in the pool is 1200 litres and is increasing.

- After what time does the volume of water first start to decrease? **(2 marks)**
- Find the volume of water in the pool when $t = 3$. **(2 marks)**
- What is the greatest volume of water in the pool? **(1 mark)**

20. Calculus, 2ADV C4 2021 HSC 27

Kenzo has a solar powered phone charger. Its power, P , can be modelled by the function

$$P(t) = 400\sin\left(\frac{\pi}{12}t\right), \quad 0 \leq t \leq 12,$$

where t is the number of hours after sunrise.

- Sketch the graph of P for $0 \leq t \leq 12$. **(2 marks)**

Power is the rate of change of energy. Hence the amount of energy, E units, generated by the solar powered phone charger from $t = a$ to $t = b$, where $0 \leq a \leq b \leq 12$ is given by

$$E = \int_a^b P(t) \, dt.$$

- Show that $E = \frac{4800}{\pi} \left(\cos \frac{a\pi}{12} - \cos \frac{b\pi}{12} \right)$. **(2 marks)**

- To make a phone call, a phone battery needs at least 300 units of energy. Kenzo woke up 3 hours after sunrise and found that his phone battery had no units of energy. He immediately began to use his solar powered charger to charge his phone battery.

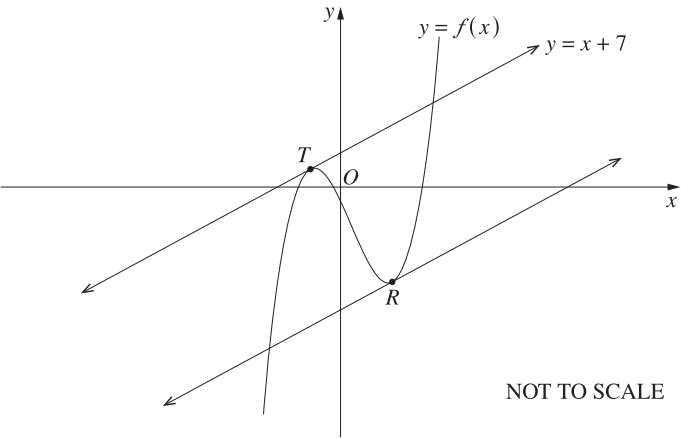
Find the least amount of time he needed to wait before he could make a phone call. Give your answer correct to the nearest minute. **(3 marks)**

- The next day, Kenzo woke up 6 hours after sunrise and again found that his phone battery had no units of energy. He immediately began to use his solar powered charger to charge his phone battery.

Would it take more time or less time or the same amount of time, compared to the answer in part (c), to charge his phone battery in order to make a phone call? Explain your answer by referring to the graph drawn in part (a). **(1 mark)**

21. Calculus, 2ADV C4 2023 HSC 28

The curve $y = f(x)$ is shown on the diagram. The equation of the tangent to the curve at point $T(-1, 6)$ is $y = x + 7$. At a point R , another tangent parallel to the tangent at T is drawn.

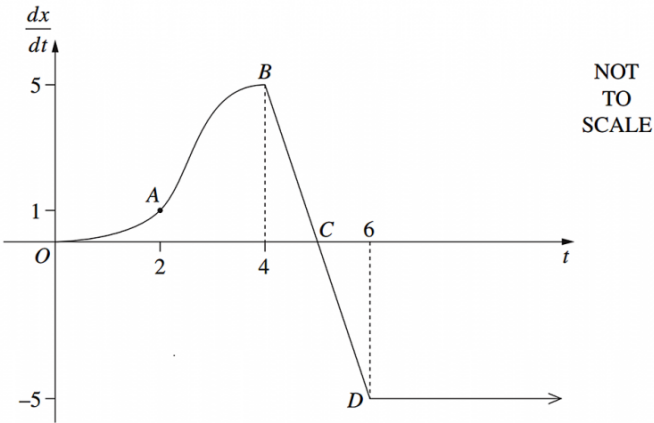


The gradient function of the curve is given by $\frac{dy}{dx} = 3x^2 - 6x - 8$.

Find the coordinates of R . (4 marks)

22. Calculus, 2ADV C4 2007* HSC 10a

An object is moving on the x -axis. The graph shows the velocity, $\frac{dx}{dt}$, of the object, as a function of time, t . The coordinates of the points shown on the graph are $A(2, 1)$, $B(4, 5)$, $C(5, 0)$ and $D(6, -5)$. The velocity is constant for $t \geq 6$.



- i. The object is initially at the origin. During which time(s) is the displacement of the object decreasing? (1 mark)
- ii. If the object travels 7 units in the first 4 seconds, estimate the time at which the object returns to the origin. Justify your answer. (2 marks)
- iii. Sketch the displacement, x , as a function of time. (2 marks)

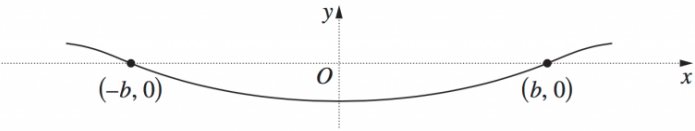
23. Calculus, 2ADV C4 2006 HSC 9b

During a storm, water flows into a 7000-litre tank at a rate of $\frac{dV}{dt}$ litres per minute, where $\frac{dV}{dt} = 120 + 26t - t^2$ and t is the time in minutes since the storm began.

- i. At what times is the tank filling at twice the initial rate? (2 marks)
- ii. Find the volume of water that has flowed into the tank since the start of the storm as a function of t . (1 mark)
- iii. Initially, the tank contains 1500 litres of water. When the storm finishes, 30 minutes after it began, the tank is overflowing.
How many litres of water have been lost? (2 marks)

24. Calculus, 2ADV C4 2008 HSC 9c

A beam is supported at $(-b, 0)$ and $(b, 0)$ as shown in the diagram.



It is known that the shape formed by the beam has equation $y = f(x)$, where $f(x)$ satisfies

$$f''(x) = k(b^2 - x^2), \quad (k \text{ is a positive constant})$$

and $f'(-b) = -f'(b)$.

i. Show that $f'(x) = k\left(b^2x - \frac{x^3}{3}\right)$. (2 marks)

ii. How far is the beam below the x -axis at $x = 0$? (2 marks)

25. Calculus, 2ADV C4 2011 HSC 9b

A tap releases liquid A into a tank at the rate of $\left(2 + \frac{t^2}{t + 1}\right)$ litres per minute, where t is time in minutes. A second tap releases liquid B into the same tank at the rate of $\left(1 + \frac{1}{t + 1}\right)$ litres per minute. The taps are opened at the same time and release the liquids into an empty tank.

i. Show that the rate of flow of liquid A is greater than the rate of flow of liquid B by t litres per minute. (1 mark)

ii. The taps are closed after 4 minutes. By how many litres is the volume of liquid A greater than the volume of liquid B in the tank when the taps are closed? (2 marks)

Worked Solutions

1. Calculus, 2ADV C4 2024 HSC 10 MC

$$A(x) = \int_0^x f(t) \, dt \text{ (note this definite integral produces a } F(x) \text{)}$$

$$A'(x) = f(x)$$

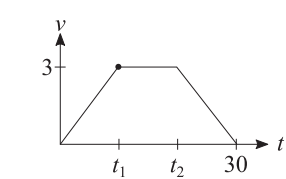
$$A''(x) = f'(x)$$

♦♦ Mean mark 41%.

POI requirements: $A''(x) = 0$ and sign (concavity) changes either side.
Inspect graph of $f(x)$ to find where $f'(x) = 0$ and gradient changes either side of possible points.
→ the three turning points all qualify
→ point Q does not qualify (gradient is positive both sides)
⇒ B

2. Calculus, 2ADV C4 SM-Bank 1 MC

Consider the velocity graph:



$t_1 \rightarrow t_2 = 10$ seconds

$\therefore$ Total distance travelled
= Area of trapezium
 $= \frac{1}{2} \times 3(10 + 30)$
 $= 60 \text{ m}$

$\Rightarrow B$

3. Calculus, 2ADV C3 2017 HSC 9 MC

Find the equation of $f'(x)$:

$m = -2, \text{ y-int} = 4$

$y = -2x + 4$

$f(x) = \int -2x + 4 \, dx$
 $= -x^2 + 4x + c$

Maximum $f(x) = 12$ when $f'(x) = 0$:

$-2x + 4 = 0$
 $x = 2$

Substitute $x = 2$ into $f(x)$:

$\therefore 12 = -2^2 + 4 \cdot 2 + c$
 $c = 8$

$\therefore f(x) = 8 + 4x - x^2$
 $\Rightarrow C$

4. Calculus, 2ADV C4 2015 HSC 9 MC

Distance travelled

$= \int_0^4 v \, dt$
=Area under the velocity curve
 $= \frac{1}{2} \times b \times h$
 $= \frac{1}{2} \times 4 \times 8$
 $= 16 \text{ metres.}$

♦♦ Mean mark 31%.

Since velocity is always positive between $t = 0$
and $t = 4$, and the original displacement = 2,
the maximum displacement = $16 + 2 = 18 \text{ metres}$
 $\Rightarrow D$

5. Calculus, 2ADV C4 2024 HSC 15

$\frac{dV}{dt} = 300 - 7.5t$

Find t when $\frac{dV}{dt} = 0$:

$300 - 7.5t = 0 \Rightarrow t = 40 \text{ hours}$

$V = \int 300 - \frac{15}{2}t \, dt$
 $= 300t - \frac{15}{4}t^2 + C$

$V = 350$ when $t = 0 \Rightarrow C = 350$

$V = 300t - \frac{15}{4}t^2 + 350$

Find V when $t = 40$:

$V = 300 \times 40 - \frac{15}{4} \times 40^2 + 350$
 $= 12\,000 - 6000 + 350$
 $= 6350 \text{ litres}$

6. Calculus, 2ADV C4 2017 HSC 13d

$$\begin{aligned}\frac{dV}{dt} &= \frac{2t}{1+t^2} \\ V &= \int \frac{2t}{1+t^2} dt \\ &= \ln (1+t^2) + c\end{aligned}$$

When $t = 0$, $V = 0$

$$\begin{aligned}0 &= \ln 1 + c \\ \therefore c &= 0\end{aligned}$$

Find V when $t = 10$:

$$\begin{aligned}V &= \ln (1+10^2) \\ &= \ln 101\end{aligned}$$

7. Calculus, 2ADV C4 2013 HSC 16a

Solution 1

$$\begin{aligned}f'(x) &= 4x - 3 \\ f(x) &= \int 4x - 3 \, dx \\ &= 2x^2 - 3x + c\end{aligned}$$

Intersection when

$$\begin{aligned}2x^2 - 3x + c &= 5x - 7 \\ 2x^2 - 8x + (7 + c) &= 0\end{aligned}$$

Since $y = 5x - 7$ is a tangent $\Rightarrow \Delta = 0$

$$\begin{aligned}b^2 - 4ac &= 0 \\ (-8)^2 - [4 \times 2 \times (7 + c)] &= 0 \\ 64 - 56 - 8c &= 0 \\ 8c &= 8 \\ c &= 1\end{aligned}$$

$$\therefore f(x) = 2x^2 - 3x + 1$$

Solution 2

$$\begin{aligned}f'(x) &= 4x - 3 \\ y = 5x - 7 \text{ (Gradient = 5)} \\ \Rightarrow 4x - 3 &= 5 \\ x &= 2\end{aligned}$$

$$\begin{aligned}f(x) &= 2x^2 - 3x + 1 \\ f(x) &\text{ passes through } (2, 3)\end{aligned}$$

$$\begin{aligned}f(2) &= 2 \times 2^2 - 3(2) + c \\ 3 &= 8 - 6 + c \\ c &= 1\end{aligned}$$

$$\therefore f(x) = 2x^2 - 3x + 1$$

8. Calculus, 2ADV C4 2019 HSC 14a

$$a = \frac{dv}{dt} = e^{2t} - 4$$

$$v = \int e^{2t} - 4 \, dt$$

$$= \frac{1}{2}e^{2t} - 4t + c$$

When $t = 0$, $v = 0$

$$0 = \frac{1}{2}e^0 - 0 + c$$

$$c = -\frac{1}{2}$$

$$\therefore v = \frac{1}{2}e^{2t} - 4t - \frac{1}{2}$$

9. Calculus, 2ADV C4 2023 HSC 13

$$P(t) = \int \frac{dP}{dt} \, dt$$

$$= \int 3000e^{2t} \, dt$$

$$= 1500e^{2t} + c$$

When $t = 0$, $P = 4000$

$$4000 = 1500e^0 + c$$

$$c = 2500$$

$$\therefore P(t) = 1500e^{2t} + 2500$$

10. Calculus, 2ADV C4 EQ-Bank 2

$$\int_0^8 28e^{0.35t} = \left[28 \times \frac{1}{0.35} e^{0.35t} \right]_0^8$$

$$= 80(e^{0.35 \times 8} - e^0)$$

$$= 80(16.44 \dots - 1)$$

$$= 1235.57 \dots$$

$$= 1240 \text{ (to 3 sig. fig.)}$$

11. Calculus, 2ADV C4 2011 HSC 4c

$$\frac{dy}{dx} = 6x - 2$$

$$y = \int 6x - 2 \, dx$$

$$= 3x^2 - 2x + c$$

Since it passes through $(-1, 4)$,

$$4 = 3(-1)^2 - 2(-1) + c$$

$$4 = 3 + 2 + c$$

$$c = -1$$

$$\therefore y = 3x^2 - 2x - 1$$

12. Calculus, 2ADV C4 2014 HSC 11f

$$f'(x) = 4x - 5$$

$$f(x) = \int 4x - 5 \, dx$$

$$= 2x^2 - 5x + C$$

Given $f(x)$ passes through $(2, 3)$:

$$3 = 2(2^2) - 5(2) + C$$

$$3 = 8 - 10 + C$$

$$C = 5$$

$$\therefore f(x) = 2x^2 - 5x + 5$$

13. Calculus, 2ADV C4 2008 HSC 5a

$$\frac{dy}{dx} = 1 - 6\sin 3x$$

$$y = \int 1 - 6\sin 3x \, dx$$

$$= x + 2\cos 3x + c$$

Passes through $(0, 7)$:

$$\Rightarrow 0 + 2\cos 0 + c = 7$$

$$2 + c = 7$$

$$c = 5$$

$$\therefore \text{Equation is } y = x + 2\cos 3x + 5$$

14. Calculus, 2ADV C4 2016 HSC 16a

i. Initial velocity when $t = 0$

$$v = 2 - \frac{4}{1} = -2 \text{ ms}^{-1}$$

ii. $v = 2 - \frac{4}{t + 1}$

$$a = \frac{dv}{dt} = \frac{4}{(t + 1)^2}$$

Particle is stationary when $v = 0$,

$$2 - \frac{4}{t + 1} = 0$$

$$2(t + 1) = 4$$
$$t = 1$$

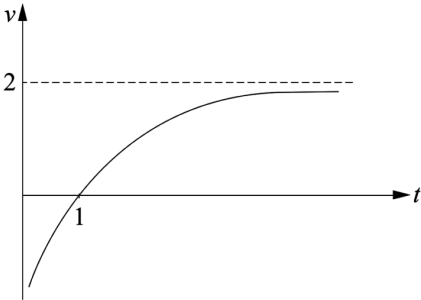
When $t = 1$,

$$\therefore a = \frac{4}{(1 + 1)^2}$$
$$= 1 \text{ ms}^{-2}$$

iii. $v = 2 - \frac{4}{t + 1}$

As $t \rightarrow \infty$, $\frac{4}{t + 1} \rightarrow 0$

$$\therefore v \rightarrow 2$$



iv. Distance travelled in 1st 7 seconds

$$= \left| \int_0^1 \left(2 - \frac{4}{t + 1} \right) dt \right| + \int_1^7 \left(2 - \frac{4}{t + 1} \right) dt$$
$$= - [2t - 4\ln(t + 1)]_0^1 + [2t - 4\ln(t + 1)]_1^7$$
$$= - [(2 - 4\ln 2) - 0] + [(14 - 4\ln 8) - (2 - 4\ln 2)]$$

Mean mark 93%.
COMMENT: Great example of low hanging fruit late in the exam for students who progress efficiently.

♦ Mean mark 47%.

♦♦ Mean mark 26%.

$$= 4\ln 2 - 2 + 12 - 4\ln 2^3 + 4\ln 2$$
$$= 10 + 8\ln 2 - 12\ln 2$$
$$= 10 - 4\ln 2 \text{ metres}$$

15. Calculus, 2ADV C4 2009 HSC 7a

i. Show $x = 2e^{-2t} + 3e^{-t} + t$

$$a = 8e^{-2t} + 3e^{-t} \quad (\text{given})$$

$$v = \int a \, dt = -4e^{-2t} - 3e^{-t} + c_1$$

When $t = 0$, $v = -6$ (given)

$$-6 = -4e^0 - 3e^0 + c_1$$

$$-6 = -7 + c_1$$

$$c_1 = 1$$

$$\therefore v = -4e^{-2t} - 3e^{-t} + 1$$

$$x = \int v \, dt$$

$$= \int (-4e^{-2t} - 3e^{-t} + 1) \, dt$$

$$= 2e^{-2t} + 3e^{-t} + t + c_2$$

When $t = 0$, $x = 5$ (given)

$$5 = 2e^0 + 3e^0 + c_2$$

$$c_2 = 0$$

$$\therefore x = 2e^{-2t} + 3e^{-t} + t \quad \dots \text{ as required}$$

ii. Particle comes to rest when $v = 0$

$$\text{i.e. } -4e^{-2t} - 3e^{-t} + 1 = 0$$

$$\text{Let } X = e^{-t} \Rightarrow X^2 = e^{-2t}$$

$$-4X^2 - 3X + 1 = 0$$

$$4X^2 + 3X - 1 = 0$$

$$(4X - 1)(X + 1) = 0$$

$$\therefore X = \frac{1}{4} \quad \text{or} \quad X = -1$$

When $X = \frac{1}{4}$:

$$e^{-t} = \frac{1}{4}$$

$$\ln e^{-t} = \ln\left(\frac{1}{4}\right)$$

$$-t = \ln\left(\frac{1}{4}\right)$$

$$t = -\ln\left(\frac{1}{4}\right) = \ln\left(\frac{1}{4}\right)^{-1} = \ln 4$$

When $X = -1$:

$$e^{-t} = -1 \quad (\text{no solution})$$

$\therefore$ The particle comes to rest when $t = \ln 4$ seconds

iii. Find x when $t = \ln 4$:

$$x = 2e^{-2t} + 3e^{-t} + t$$

$$= 2e^{-2\ln 4} + 3e^{-\ln 4} + \ln 4$$

$$= 2(e^{\ln 4})^{-2} + 3(e^{\ln 4})^{-1} + \ln 4$$

$$= 2 \times 4^{-2} + 3 \times 4^{-1} + \ln 4$$

$$= \frac{2}{16} + \frac{3}{4} + \ln 4$$

$$= \frac{7}{8} + \ln 4$$

ALGEBRA TIP: Helpful identity $e^{\ln x} = x$. Easily provable as follows:
 $e^{\ln 2} = x$
 $\Rightarrow \ln e^{\ln 2} = \ln x$
 $\Rightarrow \ln 2 = \ln x$
 $\Rightarrow x = 2$.

16. Calculus, 2ADV C4 2012 HSC 15b

i. Find v when $t = 0$:

$$\begin{aligned} v &= 1 - 2\cos 0 \\ &= 1 - 2 \\ &= -1 \end{aligned}$$

$\therefore$ Initial velocity is -1 m/s.

ii. Solution 1

Max velocity occurs when $a = \frac{dv}{dt} = 0$

$$a = 2\sin t$$

Find t when $a = 0$:

$$\begin{aligned} 2\sin t &= 0 \\ t &= 0, \pi, 2\pi, \dots \end{aligned}$$

At $t = 0$, $v = -1$ m/s

At $t = \pi$, $v = 1 - 2(-1) = 3$ m/s

$\therefore$ Maximum velocity is 3 m/s

Solution 2

$$v = 1 - 2\cos t$$

Since $-1 < \cos t < 1$

$$\begin{aligned} -2 &< 2\cos t < 2 \\ -1 &< 1 - 2\cos t < 3 \end{aligned}$$

$\therefore$ Maximum velocity is 3 m/s

iii. $x = \int v \, dt$

$$\begin{aligned} &= \int (1 - 2\cos t) \, dt \\ &= t - 2\sin t + c \end{aligned}$$

When $t = 0$, $x = 3$ (given)

$$\begin{aligned} 3 &= 0 - 2\sin 0 + c \\ c &= 3 \end{aligned}$$

$\therefore x = t - 2\sin t + 3$

iv. Find x when $v = 0$ (first time):

When $v = 0$,

$$0 = 1 - 2\cos t$$

$$\cos t = \frac{1}{2}$$

$$\begin{aligned} t &= \cos^{-1}\left(\frac{1}{2}\right) \\ &= \frac{\pi}{3} \quad (\text{first time}) \end{aligned}$$

Find x when $t = \frac{\pi}{3}$:

$$\begin{aligned} x &= \frac{\pi}{3} - 2\sin\left(\frac{\pi}{3}\right) + 3 \\ &= \frac{\pi}{3} - 2 \times \frac{\sqrt{3}}{2} + 3 \\ &= \frac{\pi}{3} - \sqrt{3} + 3 \quad \text{units} \end{aligned}$$

♦♦ Mean mark 29%
MARKER'S COMMENT: Solution 2 is more efficient here. Using the -1 and +1 limits of trig functions can be very a effective way to calculate max/min values.

♦ Mean mark 50%
MARKER'S COMMENT: Many students found $t = \frac{\pi}{3}$ but failed to gain full marks by omitting to find x . Remember that for calculus, angles are measured in radians, NOT degrees!

17. Calculus, 2ADV C4 2023 HSC 26

a. $\frac{dx}{dt} = -1.5\pi \sin\left(\frac{5\pi}{4}t\right)$

$$x(t) = -1.5\pi \int \sin\left(\frac{5\pi}{4}t\right) dt$$
$$= 1.5\pi \times \frac{4}{5\pi} \times \cos\left(\frac{5\pi}{4}t\right) + c$$
$$= \frac{6}{5} \cos\left(\frac{5\pi}{4}t\right) + c$$

When $t = 0$, $x(t) = 11.2$:

$$11.2 = \frac{6}{5} \cos(0) + c$$
$$c = 11.2 - \frac{6}{5}$$
$$= 10$$

$$x(t) = \frac{6}{5} \cos\left(\frac{5\pi}{4}t\right) + 10$$

b. Period $= \frac{2\pi}{n} = \frac{2\pi}{\frac{5\pi}{4}} = \frac{8}{5} = 1.6$ (seconds)

1st time swing reaches closest point to camera = 0.8 seconds

$$\text{Periods in next 9.2 seconds} = \frac{9.2}{1.6}$$
$$= 5.75 \text{ times}$$

$\therefore$ Swing reaches the closest point 6 times in the 1st 10 seconds

Mean mark (a) 51%.

♦♦ Mean mark (b) 31%.

18. Calculus, 2ADV C4 EQ-Bank 5

i. $m_v = -\frac{3}{5}$

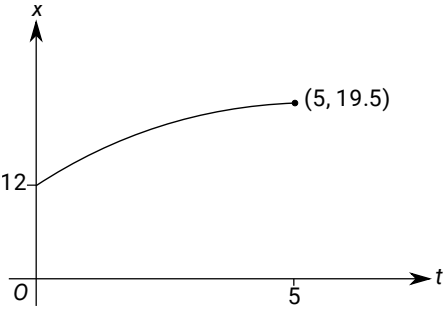
$$v = 3 - \frac{3}{5}t$$
$$x = \int v \, dt$$
$$= \int 3 - \frac{3}{5}t \, dt$$
$$= 3t - \frac{3}{10}t^2 + c$$

When $t = 0$, $x = 12 \Rightarrow c = 12$

$$\therefore x = 3t - \frac{3}{10}t^2 + 12$$

ii. When $t = 5$:

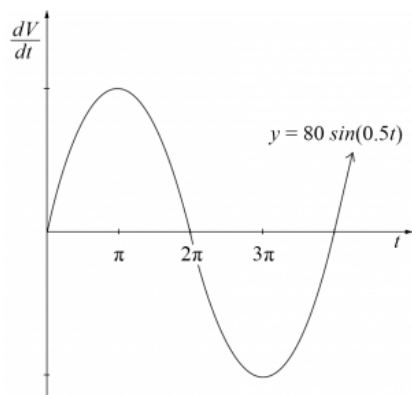
$$x = 15 - \frac{3}{10} \times 25 + 12 = 19.5$$



19. Calculus, 2ADV C4 2015 HSC 15c

i. $\frac{dV}{dt} = 80\sin(0.5t)$

Volume decreases when $\frac{dV}{dt} < 0$



♦ Mean mark (i) 37%.

$$\frac{dV}{dt} < 0 \text{ when } t > 2\pi \text{ hours}$$

$\therefore$ After 2π hours, volume starts to decrease.

ii. $V = \int \frac{dV}{dt} dt$

$$= \int 80\sin(0.5t) dt$$

$$= -160\cos(0.5t) + c$$

When $t = 0$, $V = 1200$

$$1200 = -160\cos 0 + c$$

$$c = 1360$$

$$\therefore V = -160\cos(0.5t) + 1360$$

When $t = 3$

$$V = -160\cos(0.5 \times 3) + 1360$$

$$= 1348.68 \dots \text{ litres}$$

$$= 1349 \text{ litres (nearest litre)}$$

iii. $V = -160\cos(0.5t) + 1360$

$$\Rightarrow \text{Greatest volume occurs when}$$

$$\cos(0.5t) = -1$$

$$\therefore \text{Maximum volume}$$

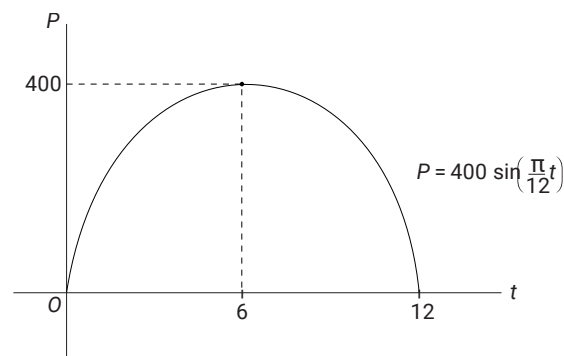
$$= -160(-1) + 1360$$

$$= 1520 \text{ litres}$$

♦♦ Mean mark (iii) 27%.

20. Calculus, 2ADV C4 2021 HSC 27

a.



b. $E = \int_a^b 400\sin\left(\frac{\pi}{12}t\right) dt$

$$= \left[-400 \cdot \frac{12}{\pi} \cos\left(\frac{\pi}{12}t\right) \right]_a^b$$

$$= -\frac{4800}{\pi} \cos\left(\frac{\pi}{12}b\right) - \left(-\frac{4800}{\pi} \cos\left(\frac{\pi}{12}a\right) \right)$$

$$= \frac{4800}{\pi} \left(\cos \frac{a\pi}{12} - \cos \frac{b\pi}{12} \right)$$

c. Find b given $E = 300$ and $a = 3$:

$$300 = \frac{4800}{\pi} \left(\cos \frac{\pi}{4} - \cos \frac{b\pi}{12} \right)$$

$$\frac{300\pi}{4800} = \frac{1}{\sqrt{2}} - \cos \frac{b\pi}{12}$$

$$\cos \frac{b\pi}{12} = \frac{1}{\sqrt{2}} - \frac{\pi}{16}$$

$$\frac{b\pi}{12} = \cos^{-1} \left(\frac{1}{\sqrt{2}} - \frac{\pi}{16} \right)$$

$$b = \frac{12}{\pi} \cos^{-1} \left(\frac{1}{\sqrt{2}} - \frac{\pi}{16} \right)$$

$$= 3.952 \dots$$

$$= 3 \text{ h } 57 \text{ m (nearest minute)}$$

$\therefore$ Least time before phone is charged = 57 minutes

d. The power produced is at its peak when $t = 6$.

$\therefore$ It will charge in less time.

♦♦ Mean mark part (c) 29%
COMMENT: It is arguable that the nearest minute may also be 3h 58 m as the phone is not adequately charged at 3h 57 m.

♦♦ Mean mark part (d) 22%

21. Calculus, 2ADV C4 2023 HSC 28

Gradient at $R = 1$:

$$3x^2 - 6x - 8 = 1$$

$$3x^2 - 6x - 9 = 0$$

$$3(x^2 - 2x - 3) = 0$$

$$3(x + 1)(x - 3) = 0$$

x -coordinate of $R = 3$

$$y = \int 3x^2 - 6x - 8 \, dx$$

$$= x^3 - 3x^2 - 8x + c$$

Graph passes through $(-1, 6)$:

$$6 = -1 - 3 + 8 + c$$

$$c = 2$$

$$y = x^3 - 3x^2 - 8x + 2$$

When $x = 3$:

$$y = 27 - 27 - 24 + 2 = -22$$

$$\therefore R(3, -22)$$

◆ Mean mark 51%.

22. Calculus, 2ADV C4 2007* HSC 10a

i. Displacement is reducing when the velocity is negative.

$$\therefore t > 5 \text{ seconds}$$

ii. At B , the displacement = 7 units

Considering displacement from B to D :

Since the area below the graph from

B to C equals the area above the

graph from C to D , there is no change

in displacement from B to D .

Considering $t \geq 6$

Time required to return to origin

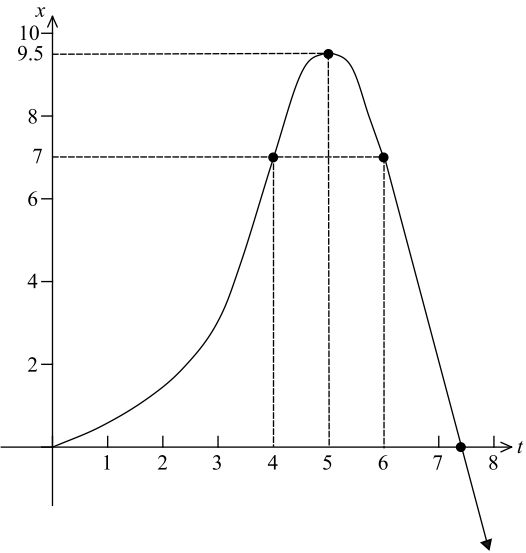
$$t = \frac{d}{v}$$

$$= \frac{7}{5}$$

$$= 1.4 \text{ seconds}$$

$\therefore$ The particle returns to the origin after 7.4 seconds.

iii.



23. Calculus, 2ADV C4 2006 HSC 9b

i. $\frac{dV}{dt} = 120 + 26t - t^2$

When $t = 0$, $\frac{dV}{dt} = 120$

Find t when $\frac{dV}{dt} = 240$

$$240 = 120 + 26t - t^2$$

$$t^2 - 26t + 120 = 0$$

$$(t - 6)(t - 20) = 0$$

$$t = 6 \text{ or } 20$$

$\therefore$ The tank is filling at twice the initial rate
when $t = 6$ and $t = 20$ minutes

ii. $V = \int \frac{dV}{dt} dt$

$$= \int 120 + 26t - t^2 dt$$

$$= 120t + 13t^2 - \frac{1}{3}t^3 + c$$

When $t = 0$, $V = 0$

$$\Rightarrow c = 0$$

$$\therefore V = 120t + 13t^2 - \frac{1}{3}t^3$$

iii. Storm water volume into the tank when $t = 30$

$$= 120(30) + 13(30^2) - \frac{1}{3} \times 30^3$$

$$= 3600 + 11\,700 - 9000$$

$$= 6300 \text{ litres}$$

Total volume = $6300 + 1500$

$$= 7800 \text{ litres}$$

$$\therefore \text{Overflow} = 7800 - 7000$$

$$= 800 \text{ litres}$$

24. Calculus, 2ADV C4 2008 HSC 9c

i. Show $f'(x) = k\left(b^2x - \frac{x^3}{3}\right)$

$$f''(x) = k(b^2 - x^2)$$

$$f'(x) = \int k(b^2 - x^2) dx$$

$$= k \int b^2 - x^2 dx$$

$$= k\left(b^2x - \frac{x^3}{3}\right) + c$$

Since S.P. exists at $x = 0$

$$\Rightarrow f'(x) = 0 \text{ when } x = 0$$

$$0 = k(b^2 \cdot 0 - 0) + c$$

$$c = 0$$

$$\therefore f'(x) = k\left(b^2x - \frac{x^3}{3}\right) \dots \text{as required}$$

ii. $f(x) = \int f'(x) dx$

$$= k \int b^2x - \frac{x^3}{3} dx$$

$$= k\left(\frac{b^2x^2}{2} - \frac{x^4}{12}\right) + c$$

We know $f(x) = 0$ when $x = b$

$$\Rightarrow 0 = k\left(\frac{b^2 \cdot b^2}{2} - \frac{b^4}{12}\right) + c$$

$$c = -k\left(\frac{6b^4}{12} - \frac{b^4}{12}\right)$$

$$= -k\left(\frac{5b^4}{12}\right)$$

$$= -\frac{5kb^4}{12}$$

$$\therefore \text{When } x = 0, \text{ the beam is } \frac{5kb^4}{12} \text{ units}$$

below the x -axis.

25. Calculus, 2ADV C4 2011 HSC 9b

i. Show difference in flow rate $(D) = t$

$$\begin{aligned} D &= \left(2 + \frac{t^2}{t+1} \right) - \left(1 + \frac{1}{t+1} \right) \\ &= \frac{2(t+1) + t^2}{t+1} - \frac{(t+1) + 1}{t+1} \\ &= \frac{2t + 2 + t^2 - t - 2}{t+1} \\ &= \frac{t^2 + t}{t+1} \\ &= \frac{t(t+1)}{t+1} \\ &= t \quad \dots \text{ as required} \end{aligned}$$

◆◆◆ Mean mark 16%

MARKER'S COMMENT: Many students incorrectly differentiated in part (i). The 1 mark allocation indicates the answer will not require an involved multi-step process.

ii. Difference in Volume

$$\begin{aligned} &= \int_0^4 \left(2 + \frac{t^2}{1+t} \right) dt - \int_0^4 \left(1 + \frac{t}{1+t} \right) dt \\ &= \int_0^4 t dt \quad (\text{using part(i)}) \\ &= \left[\frac{t^2}{2} \right]_0^4 \\ &= \frac{16}{2} - 0 \\ &= 8 \end{aligned}$$

◆◆◆ Mean mark 15%

MARKER'S COMMENT: Few students were able to answer this part. Previous parts of any question should be front and centre of your thinking when working out strategies.

$\therefore$ There is 8 litres more of liquid A than B .